RUIQI HE

CONTACT

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https://ricky-s-portfolio-olive.vercel.app/

EDUCATION

The Ohio State University
Columbus, OH
Bachelor of Science
ASC Honors College
Major: Data Analytics
Specialization: Biomedical and Public Health Analytics
Minor: Business
Dean's List
Expected Graduation: May 2028

West Chester East High School
West Chester, PA

High School Diploma
June,2024
Honors Roll every semester
Pennsylvania State Academic Team Competition Champion

SKILLS

- Programming Languages: Java, Python, R and C++
- Libraries & Tools: Pandas, scikit-learn, Random Forest, TensorFlow, PyTorch, LangChain, LLM, ML models
- Software: Proficient in Microsoft Office applications (Word, Excel, PowerPoint, etc.)

LANGUAGES

- English: Native
- Chinese: Native
- Spanish: Basic proficiency

EXPERIENCE

Undergraduate Research Intern – AIMed Lab | Summer 2026 – Present | The Ohio State University | Department of Biomedical
Mentor: Dr. Ping Zhang

- Competing in the 2026 George B. Moody PhysioNet Challenge to develop predictive models that forecast cognitive impairment diagnoses from polysomnography (sleep study) data
- Engineering ML pipelines to analyze multi-modal physiological signals including EEG, EOG, EMG, ECG, respiratory effort, and airflow to identify patients at risk for mild cognitive impairment, Alzheimer's disease, and dementia
- Processing large-scale PSG datasets (1.2TB training set across 6,600+ recordings) with heterogeneous channel availability, variable sampling rates, and real-world noise
- Optimizing age-conditioned AUROC scoring metrics to predict diagnoses occurring 1-6 years post-sleep study

PROJECTS

ASA DataFest @ OSU | Best Visualization Award | Spring 2026
Competition Project for Stormont Vail Health

- Engineered a dynamic geospatial dashboard with regional health trend identification, lasso-based multi-region selection, department-level filters, and top-diagnosis frequency analysis
- Leveraged NVIDIA RAPIDS cuDF on the DGX Spark to accelerate data preprocessing and cleaning, significantly improving dashboard responsiveness for real-time filtering across 100K+ patient encounters
- Extracted actionable insights for hospital administration on patient distribution patterns and healthcare access barriers across Kansas, enabling data-driven resource allocation decisions
- Demo: https://www.youtube.com/watch?v=kBPFNHTOf_g&t=4

CLARA (Clairvoyant Loss Avoidance & Risk Advisor) | Hackathon Win | Spring 2026

- Built a prototype AI-driven enterprise risk management platform for investors and portfolio managers
- Implemented real-time VaR & Expected Shortfall monitoring, Monte Carlo simulations, and hedge optimization concepts
- Automated 10-K risk extraction, regime detection, and stress testing for demonstration
- Leveraged Gemini for financial reasoning and Watson AI for conversational insights

NYC Housing Sales Analytics Website | Hackathon Winner | Spring 2026

- Built a full-stack analytics website analyzing 34,394 NYC property sales using Python + Databricks
- Created an end-to-end data pipeline with geospatial borough assignment and residential filtering
- Developed an interactive dashboard with ML-based price prediction and geospatial insights
- Live demo: vision-housing-ai.lovable.app

Doctor-in-the-Loop AI System | 1st Place — Foundation of AI Track | Autumn 2025

- Developing an explainable AI system to analyze X-rays and suggest editable treatment plans
- Generates overlay explanations for findings to assist doctors' decision-making
- Won 1st Place at Research Gala, evaluated by a panel of judges

Talkora — Gemini Hackathon | Spring 2026

- Built a real-time voice-based language learning app using Gemini Live API
- Implemented adaptive CEFR difficulty, AI personas, and live pronunciation/grammar feedback
- Engineered a low-latency browser-to-API audio pipeline using Web Audio API

AI Study Planner | Top Project — BDAA × Lovable Hackathon | Autumn 2025

- Built an AI-powered platform to generate personalized study schedules and exam prep plans
- Implemented note summarization from course content, user notes, and audio recordings
- Included a quiz feature to track and test student progress

HONORS

- National Buckeye Scholar
- Trustees Scholarship

ACTIVITIES

- BDAA(Big Data & Analytics Association): Autumn 2025 - Present
  - Foundation of AI TA: Spring 2026 - Present
- Formula Buckeye: Autumn 2025 - Present
- Academic Team: Autumn 2022 – Spring 2024
- Mock Trial: Autumn 2022 – Spring 2024